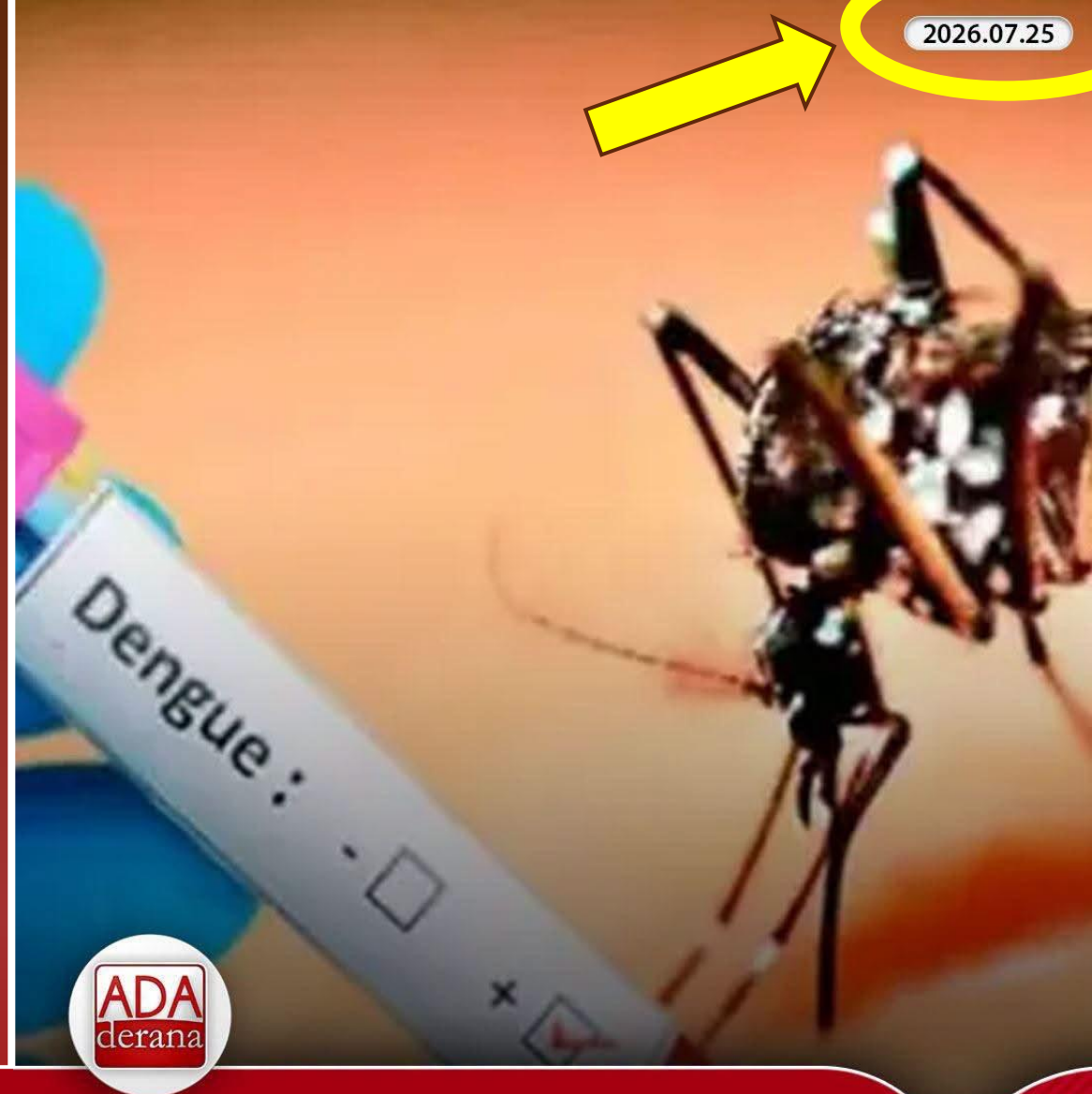


Dengue Patient Monitoring System



TEAM NEXORA



Sri Lanka's dengue death toll rises to 61
as cases exceed 80,000



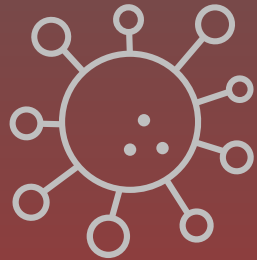


The Danger of Dengue

CURRENT SITUATION OF SRI LANKA



Reported Cases So Far



80,000+

***Cases so
for This
Year***

5,000+

Last Week

61

***Deaths so
for This Year***



How patients are monitored

- Body Temperature
- Pulse Rate , Pulse Volume
- Respiratory Rate
- Blood Pressure
- Pulse Pressure
- CRFT
- PCV
- Platelet Count
- WBC
- IP/OP





Problem We Identified

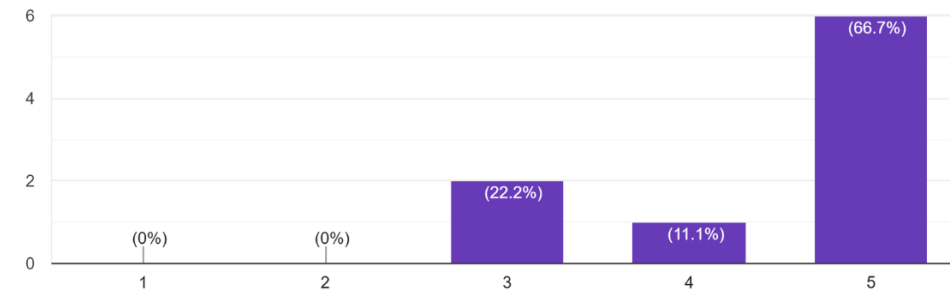
Manual observations and
periodic laboratory tests

Require significant time, effort,
and human resources

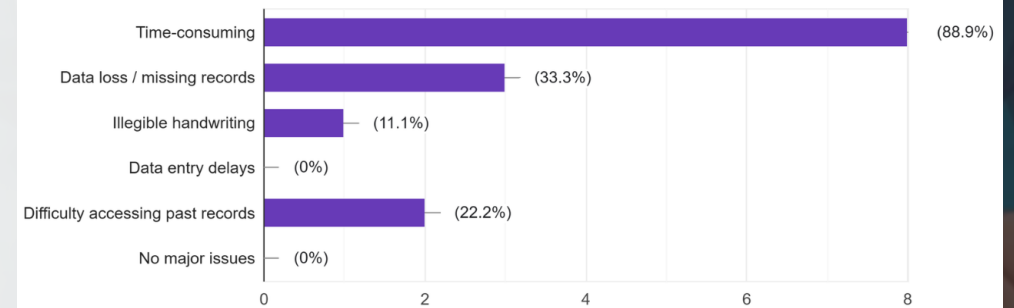
Existing manual data recording
approach is not suitable

Survey Data

Would automatic digital data recording be beneficial? (Linear scale 1–5)



What problems do you face with the current data recording method?





Smart Dengue Patient Monitoring System

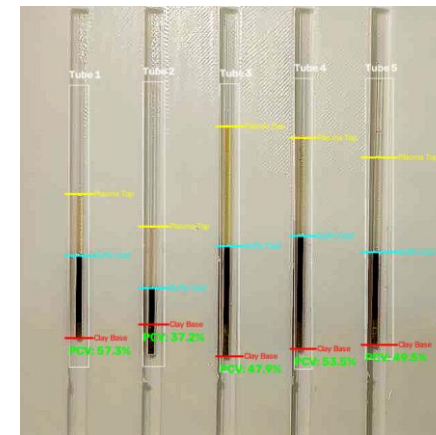
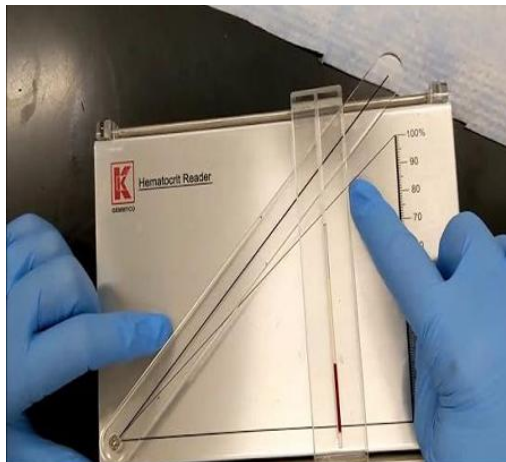
- Pulse Rate , Respiratory Rate
- Blood Pressure
- CRFT
- PCV
- IP/OP

OUR PRODUCTS

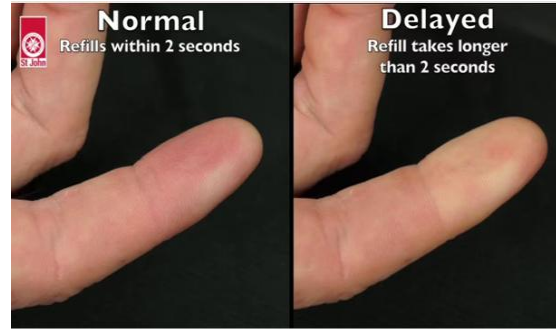
Blood Pressure



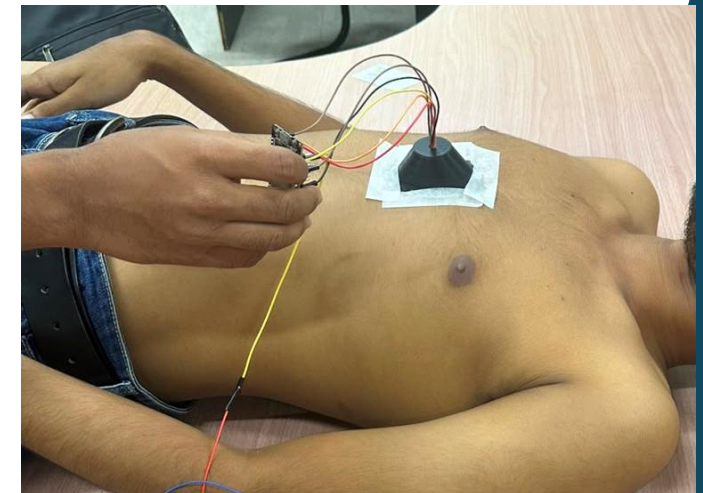
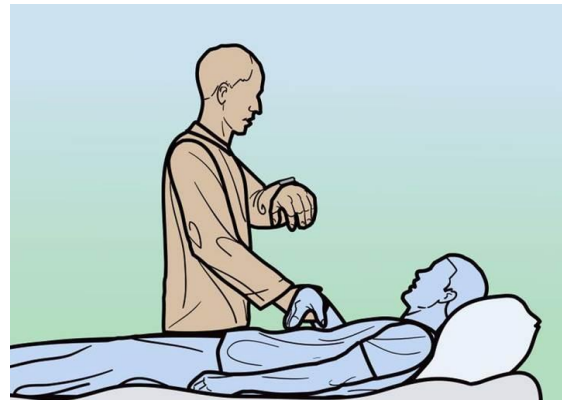
PCV



CRFT



Pulse Rate & Respiratory rate



URINE OUTPUT



Web Dashboard

[illegible][illegible]

Solution Validation



**National Institute of
Infectious Diseases
(IDH) | Angoda**



**Sri Jayewardenepura
General Hospital**

Medical Device Classification

- **Intended Use:** Intended as a non-invasive multi-parameter patient monitoring and clinical decision support system.
- **Risk Category:** Expected to fall under a moderate-risk medical device (e.g., Class II/IIa depending on jurisdiction and intended use), subject to confirmation by the relevant regulatory authority.
- **Determining Factors:** Final classification will depend on clinical claims, intended use, software functionality, and level of influence on clinical decision making.



Regulatory Framework



- **ISO 13485** – Quality Management System (QMS) for medical devices.
- **ISO 14971** – Risk management throughout the product life cycle.
- **IEC 62304** – Medical device software life cycle processes.
- **IEC 60601 Series** – Electrical safety and electromagnetic compatibility.
- **IEC 62366** – Usability engineering for safe human-device interaction.
- **ISO 27001 (planned)** – Information security management for cloud infrastructure.
- Compliance with applicable patient data protection and privacy regulations.

Risk Management

Risk management is incorporated throughout the development process following ISO 14971.

- **Hazard identification and risk analysis.**
- **Software verification and validation.**
- **Hardware reliability and electrical safety testing.**
- **Cybersecurity and secure communication.**
- **Patient data integrity and traceability.**
- **Human factors and usability evaluation**



Intellectual Property Strategy

- **Patent landscape review to identify existing technologies.**
- **Freedom-to-Operate (FTO) assessment to minimize infringement risks.**
- **Planned provisional patent filing covering:**
 - Machine vision-based multi-capillary PCV measurement.
 - Automated capillary refill time measurement mechanism.
 - Automated urine output and colour monitoring system.
 - AI-assisted physiological parameter estimation.
 - Integrated multi-parameter patient monitoring platform.
 - Digital health ecosystem architecture and workflow.
- **Future Patent Cooperation Treaty (PCT) application for international protection.**



SDG Relevance



- Enables timely detection of dengue deterioration
- Continuous vital monitoring reduces preventable deaths
- Reduces nurse workload enabling better patient care
- Digital records support faster clinical decision-making



- Integrates AIoT into hospital infrastructure
- Drives innovation in smart medical devices
- Builds digital health infrastructure in LMICs



- Eliminates paper-based monitoring - reduces carbon footprint
- Addresses climate-driven disease outbreak risks
- Strengthens healthcare system resilience

Industry 5.0

Human machine collaboration in patient care

– Machine does –

- Continuously monitors heart rate
- Tracks respiratory rate in real time
- Estimates blood pressure non-invasively
- Flags early warning signs automatically
- Visualizes trends on nurse dashboard



– Clinician does –



- Enters clinical observations and assessments
- Makes all treatment and intervention decisions
- Reviews and validates system alerts
- Maintains full control over patient care
- Places the patient at the center of care

Market Analysis

Sri Lanka

- Over 73,000 annual dengue cases
- Approximately 400 daily hospital admissions
- 2,800 patients in-hospital per week
- Home monitoring market Opportunity

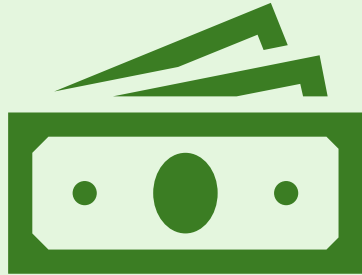


Global

DAILY HOSPITAL PATIENT VOLUME — REGIONAL COMPARISON

India	800 patients/day	800 / day
Bangladesh	700 patients/day	700 / day
Sri Lanka	400 patients/day	400 / day
Pakistan	400 patients/day	400 / day

Revenue Streams



80%
ROI

Dengue Patient Monitoring System (Hardware)

Hardware Module	Estimated Mfg. Cost	Unit Price
Base Unit	\$70	\$120
PCV Module	\$30	\$50
CRFT Module	\$15	\$30
Heart Rate Module	\$20	\$40
Blood Pressure Module	\$15	\$30
Urine Output Module	\$25	\$45
	= \$175	= \$315
Category		Market Price
Market price on one fully loaded unit(Gov)		\$250
Market price on one fully loaded unit(Private)		\$315

Annual Service & Maintenance Operations (Software)

Category	Market Price
Initial Setup & Licensing (Upon Purchase)	\$150
Annual Maintenance Contract (Gov)	\$50 / unit
Annual Maintenance Contract (Private)	\$100 / unit

5

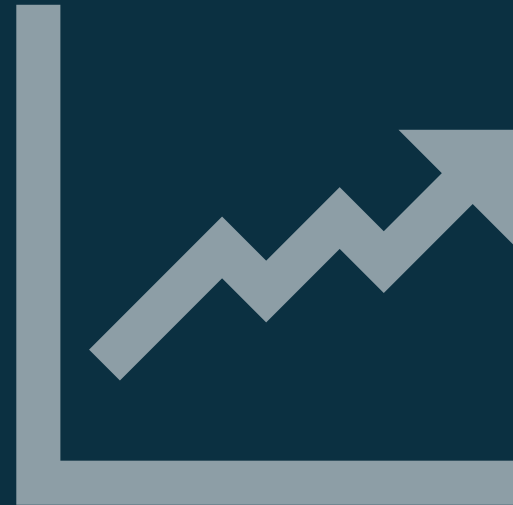
4

3

2

1

SCALABILITY



5

4

3

2

01

Other Diseases

- ***Critical Care /ICU Patients***
- ***Post-Surgery Monitoring***
- ***Heart Failure Patients***
- ***Malaria Monitoring***

1

5

4

3

02

Home Care Kit

***Enabling patients to
receive continuous health
monitoring within their
residences.***

2

1

5

4

03

From Ward to Hospital

***Centralized Hospital
Management System***

3

2

1

5

04

To Doctors and General Patients

- Customized data recording system for doctors for their personal use.
- Customized profile for general patients for their personal use.

4

3

2

1

05

***WHY TO BUY
OUR
PRODUCT***



5

4

3

2

1

Q & A

Scan



For more Project Details

